

- 9 (a) Most carnivorous mammals need to move to hunt their prey.

Outline why a carnivorous mammal makes more use of its nervous system, rather than its endocrine system, when it hunts.

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- (b) Fig. 9.1 is a diagram of a motor neurone.

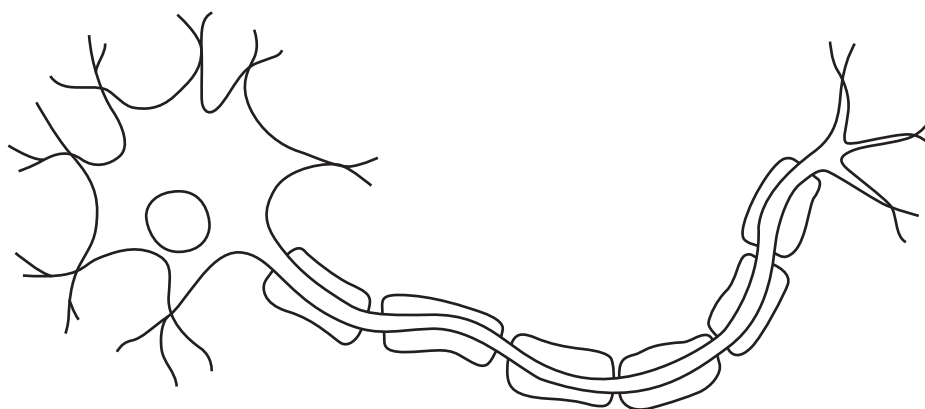


Fig. 9.1

On Fig. 9.1, add label lines and the letters **R**, **S** and **T** to label a part of the neurone that:

- can become depolarised – use the letter **R**
- contains many mitochondria – use the letter **S**
- acts as an insulator – use the letter **T**.

[3]

(c) Fig. 9.2 summarises changes that occur during the contraction of a sarcomere.

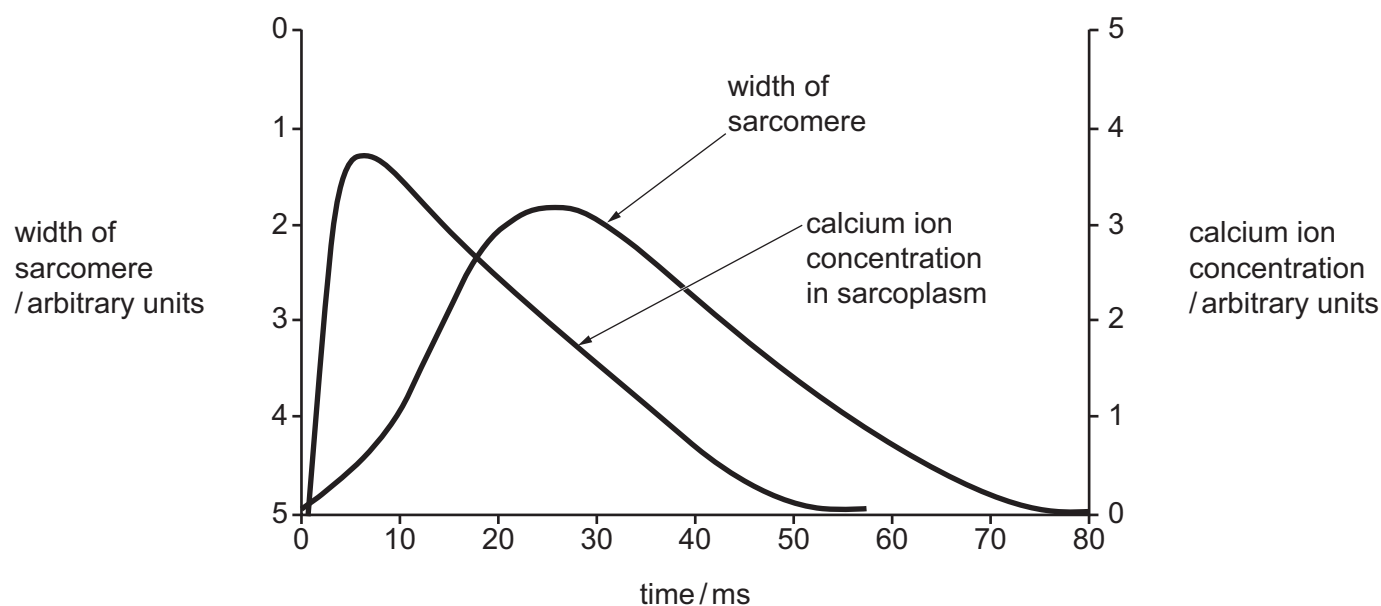


Fig. 9.2

- (i) Suggest an explanation for the shape of the curve that shows changes in the width of the sarcomere.

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- (ii) Suggest an explanation for the curve that shows changes in calcium ion concentration.

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